IMAGE PROCESSING

1. Component	Role	Nature
Gemini (vision-language model)	Looks at the image and reports what it sees	Probabilistic, interprets pixels
Python rules layer	Turns that report into a verdict	Deterministic, fixed if/else logic

2. How Gemini "looks" at the photo

Your prompt tells Gemini:

1. Report only what is visibly in the image.
2. Do not guess hidden things (wiring inside walls, earthing, circuit capacity).
3. Pick hazards from a fixed list: **water_or_dampness, exposed_or_damaged_wiring, combustibles_or_fire_hazard, pedestrian_or_cable_obstruction, cramped_or_inaccessible, visible_electrical_hazard, none_visible, uncertain.**
4. Return JSON only, with image quality, hazards (type, severity, confidence), summary, and limitations.

3. Worked example

Photo	Gemini says	Rules layer result
A	Clear image, no hazards, none_visible	CONTINUE_TO_ELECTRICIAN
B	Clear image, cable lying across walkway, medium confidence	CORRECTION_REQUIRED
C	Dark, blurry image	RETAKE_PHOTO

CHATBOT

1. EV-only scope: Answers questions about EV charging, chargers, cables, charging errors and charging safety

2. Validate: validContents() checks that the conversation has an acceptable structure, contains only user/model roles and stays within configured message and text limits.

PROMPTS

1. Role: Act as ChargeSathi, an EV charging assistant for Indian EV owners, covering charging problems, connectors, chargers and safety.

2. Language & format: Reply in the user's language, classify severity as [[SEV:low]], [[SEV:medium]] or [[SEV:high]], and keep responses under approximately 150 words.

3. Safety rules: For emergencies, give immediate safety instructions. Never recommend bypassing MCB/RCCB protection, opening electrical panels or repairing damaged cables.

4. Troubleshooting & scope: Give the likely cause, 3–6 safe troubleshooting steps and a next step. Answer only EV-charging-related questions and do not invent error-code meanings.

5. Technical detail: The prompt is sent to Gemini through systemInstruction in the backend request. The model uses temperature: 0.3 to encourage more consistent responses